

Selection as Calibrated Symmetrization

May 2026

1 Point of the note

The selection principle in Plan 1 looks like a penalized minimization, but it has a useful geometric interpretation. It is an implicit, constrained symmetrization step:

replace Ω_0 by the best torsion-energy representative with essentially the same α -asymmetry.

It is not Schwarz or Steiner symmetrization. Those flows drive the asymmetry to zero, while the selection principle must preserve enough asymmetry to keep a near-counterexample alive. The correct object is therefore a *calibrated symmetrization*: decrease the torsion deficit and volume error, but constrain the smooth asymmetry level.

This point of view also explains why the perturbing functional can be taken to be Lipschitz, with uniformly bounded boundary Lipschitz constant.

2 The selected resolvent

Let $\Omega_0 \subset B_R$, $|\Omega_0| = |B_1|$, and set

$$\varepsilon := \alpha(\Omega_0), \quad \delta := E(\Omega_0) - E(B_1), \quad q := \delta/\varepsilon.$$

Recall the BDV smooth asymmetry

$$\alpha(U) = \int_{U \Delta B_1(x_U)} |1 - |x - x_U|| \, dx,$$

where x_U is the barycenter. On bounded sets,

$$|\alpha(U) - \alpha(V)| \leq C_R |U \Delta V|. \quad (2.1)$$

For a parameter $\tau \ll 1$, define

$$\Phi_{\varepsilon, \tau}(s) := \sqrt{\varepsilon^2 + \tau^2(s - \varepsilon)^2}.$$

Then

$$0 \leq \Phi'_{\varepsilon, \tau}(s) \leq \tau \quad \text{in absolute value.} \quad (2.2)$$

The selected set is

$$S_\tau(\Omega_0) \in \operatorname{argmin}_{U \subset B_R} \{E(U) + f_{\hat{\eta}}(|U|) + \Phi_{\varepsilon, \tau}(\alpha(U))\}. \quad (2.3)$$

This is exactly the Plan 1/BDV penalized minimizer, with the sequence parameter replaced by the single parameter τ .

Proposition 2.1 (Calibrated symmetrization properties). *Let $U = S_\tau(\Omega_0)$, and let*

$$F_{\hat{\eta}}(U) := E(U) + f_{\hat{\eta}}(|U|).$$

Then

$$F_{\hat{\eta}}(U) \leq F_{\hat{\eta}}(\Omega_0). \quad (2.4)$$

Moreover

$$\frac{|\alpha(U) - \varepsilon|}{\varepsilon} \leq \frac{\sqrt{2q + q^2}}{\tau} + O_R(q), \quad (2.5)$$

with the harmless $O_R(q)$ coming from the usual volume-penalty normalization.

Proof. Minimality gives

$$F_{\hat{\eta}}(U) + \Phi_{\varepsilon, \tau}(\alpha(U)) \leq F_{\hat{\eta}}(\Omega_0) + \Phi_{\varepsilon, \tau}(\varepsilon) = F_{\hat{\eta}}(\Omega_0) + \varepsilon.$$

Since $\Phi_{\varepsilon, \tau} \geq \varepsilon$, this implies (2.4). Also, using the lower bound $F_{\hat{\eta}}(U) \geq F_{\hat{\eta}}(B_1) - O_R(\delta)$ supplied by the BDV volume-penalty lemma, the same inequality gives

$$\sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2} \leq \varepsilon + \delta + O_R(\delta\varepsilon).$$

Squaring and dividing by ε^2 gives (2.5). $\square$

Thus the map S_τ really is an improvement step: it lowers the volume-penalized torsion energy, while the soft constraint prevents the asymmetry from collapsing. With the canonical choice $\tau = q^{1/4}$, the relative asymmetry loss is $O(q^{1/4})$.

3 The flow behind the minimization

For a smooth deformation with normal velocity V , the torsion energy has the shape derivative

$$E'(U)[V] = -\frac{1}{2} \int_{\partial U} |\nabla u_U|^2 V \, d\mathcal{H}^{N-1}. \quad (3.1)$$

The volume term contributes $f'_{\hat{\eta}}(|U|) \int_{\partial U} V \, d\mathcal{H}^{N-1}$. The smooth asymmetry has first variation

$$\alpha'(U)[V] = \int_{\partial U} \Psi_U V \, d\mathcal{H}^{N-1}, \quad (3.2)$$

where, up to an additive constant absorbed into the volume multiplier,

$$A_U := \frac{1}{|U|} \int_U \frac{z - x_U}{|z - x_U|} dz, \quad \Psi_U(x) := |x - x_U| - A_U \cdot (x - x_U). \quad (3.3)$$

Since $|A_U| \leq 1$, the bracket is globally Lipschitz:

$$\text{Lip}(\Psi_U) \leq 2. \quad (3.4)$$

At a smooth selected minimizer, stationarity gives the Bernoulli law

$$|\nabla u_U(x)|^2 = \Lambda + a \Psi_U(x), \quad x \in \partial U, \quad (3.5)$$

where

$$a = 2\Phi'_{\varepsilon, \tau}(\alpha(U)), \quad |a| \leq 2\tau, \quad (3.6)$$

up to the harmless convention-dependent factor 2. Hence the boundary coefficient in (3.5) has a Lipschitz constant bounded by $C\tau$. Once the free boundary lies in a fixed annulus around the selected center, Ψ_U has C^k bounds depending only on N, R, k , because the only nonlinear term is $|x - x_U|$.

This is the flow interpretation: the formal normal velocity

$$V = \frac{1}{2} |\nabla u_U|^2 - f'_{\tilde{\eta}}(|U|) - \Phi'_{\varepsilon, \tau}(\alpha(U)) \Psi_U \quad (3.7)$$

is a constrained torsion-symmetrizing descent. The selected minimizer is a stationary point of this descent. The direct minimization is an implicit Euler/resolvent version of the flow, and it avoids having to construct the flow itself.

4 Local replacement version

The same idea can be phrased without global minimization. Given a current positivity set $U = \{u > 0\}$, choose a ball $B \subset B_R$, keep u fixed outside B , and replace u inside B by a competitor v that lowers

$$\frac{1}{2} \int |\nabla v|^2 - \int v + f_{\tilde{\eta}}(|\{v > 0\}|) + \Phi_{\varepsilon, \tau}(\alpha(\{v > 0\})).$$

Because of (2.1)–(2.2), the last term changes by at most

$$C_R \tau |\{u > 0\} \Delta \{v > 0\}|. \quad (4.1)$$

Therefore any limit of a descent-by-local-replacements procedure satisfies the one-phase quasi-minimality inequality

$$\begin{aligned} \frac{1}{2} \int |\nabla u|^2 - \int u + f_{\tilde{\eta}}(|\{u > 0\}|) &\leq \frac{1}{2} \int |\nabla v|^2 - \int v + f_{\tilde{\eta}}(|\{v > 0\}|) \\ &\quad + C_R \tau |\{u > 0\} \Delta \{v > 0\}|. \end{aligned} \quad (4.2)$$

This is precisely the Alt–Caffarelli/BDV input used for density, nondegeneracy, Lipschitz control of u , and eventual free-boundary regularity. In this local form the selection step looks like a smoothing algorithm: keep making torsion-improving replacements, but charge any asymmetry drift by a uniformly Lipschitz cost.

5 What this does and does not replace

1. It does replace the vague compactness language by a concrete “symmetrizing resolvent”:

$$\Omega_0 \mapsto S_{\tau}(\Omega_0).$$

The map lowers volume-penalized torsion energy and keeps the smooth asymmetry within a controlled factor.

2. It explains why the extra functional can be chosen with bounded Lipschitz constant. The scalar penalty has slope at most τ , and the asymmetry bracket has globally bounded Lipschitz norm on B_R . On the eventual annular collar the bracket has all C^k bounds needed for the Schauder bootstrap.

3. It is not a literal Schwarz/Steiner symmetrization. Such symmetrizations improve the energy by moving toward the ball, but they also erase the asymmetry. Selection must improve the set while keeping it at the same asymmetry level, because otherwise a near-counterexample disappears.
4. It does not by itself prove graph entry. It gives a regular quasi-minimizer with a controlled Bernoulli coefficient. The later Plan 1 steps still need density, Hausdorff-to-flatness, and patching to a global graph.

6 Practical takeaway

The underlying structure of the Plan 1 selection step is:

torsion symmetrization constrained to the asymmetry level of the input

implemented by an implicit variational step. For the purpose of regularity, the only property of the asymmetry constraint that matters is

$$|\Phi_{\varepsilon,\tau}(\alpha(U)) - \Phi_{\varepsilon,\tau}(\alpha(V))| \leq C_{R\tau} |U\Delta V|,$$

and, at a smooth boundary,

$$|\nabla u_U|^2 = \Lambda + a\Psi_U, \quad |a| \leq C\tau, \quad \text{Lip}(\Psi_U) \leq C.$$

So yes: after selection, one may treat the added functional as a Lipschitz boundary forcing with bounded Lipschitz constant. The penalized minimization is the rigorous surrogate for the desired smoothing/flow-like symmetrization.